

 Ingredion (Thailand) Co., Ltd.		SAMSON VALVE SPECIFICATION					
		PROJECT SPRINT 18K TPA SPRAY DRYER					
1	Customer	AEC INDUSTRIAL ENGINEERING CO., LTD.				Tag No.	PV-60201
2	Project No.					Item	1 Qty. 1 Status
3	Unit					Quot. No.	Revision 0
4	PID	325-01-600-PD-02				Order No.	Date 26.01.2026
5	Line No.						
6	PROCESS DATA						
7	Service					Medium	Steam
8	Phase	vapor				Medium Cust.	Steam (20 Bar G)
9			Unit	Case 1	Case 2	Case 3	
10	Flow Rate	W	kg/h	4000	6500	7000	
11	Inlet Pressure	p1	bar(g)	16	18	20	
12	Outlet Pressure	p2	bar(g)	15	17.5	18	
13	Inlet Temperature	T1	°C	206	214	240	
14	Spec. Volume	v1	m³/kg	0.117	0.106	0.103	
15	Vapor Pressure	pv	bar(a)				
16	Critical Pressure	pc	bar(a)				
17	Ratio of Specific Heats	γ	-	1.28	1.28	1.29	
18	Compressibility	Z	-				
19	Inlet Viscosity	η	mPas	0.0159	0.0162	0.0174	
20	Outlet Velocity	w	Mach	0.0368	0.0518	0.0564	
21	Valve Coeff. Calculated	Cv	-	51.7	112	61.1	
22	Relative Travel	T	%	56.8	78.9	60.7	
23	SPL(SAMSON Standard values)	LA	dB (A)	56	52	69	
24	DESIGN			77	LINE		
25	Shut-Off Pressure Min	0 bar(g)	Max 24 bar(g)	78	Size / Rating / Pipe Class		
26	t1max		Max 240 °C	79	In NPS 6" / Schedule 40 /		
27	Supply Pressure Min	4 bar(g)	Max bar(g)	80	Out NPS 6" / Schedule 40 /		
28	Pressure Min	bar(a)	Max bar(a)	81	Pipe Inner Dia. / Wallthickness 154,1 mm / 7,1 mm		
29	Temperature Min	°C	Max 250 °C	82	Insu. none		
30	Ambient Temp. Min	°C	Max °C	83	POSITIONER		
31	VALVE BODY / BONNET			84	Manufacturer / Type SAMSON / 3730-1		
32	Manufacturer / Type	SAMSON / 3241		85	Device Type TROVIS 3730-1		
33	Style	Globe		86	Quantity 1		
34	Nominal Size	NPS 4		87	Material Aluminium		
35	Rating	Class 300		88	Elec. / Pneu. Conn. / Gauge block G1/4, alu		
36	Body Material	A351 CF8M		89	Input Sig. / Commu. 4 to 20 mA / -		
37	Connection In/Out	flanges		90	Ex Approval / Cert. T3730-1-110 ATEX ia / ATEX		
38	Connection In/Out	RF (Ra 3.2 - 6.3), ASME B16.5		91	AIR SET		
39	Bonnet Type	Insulating section		92	Manufacturer / Type SMC / AW		
40	Bellow Material	-		93	Device Type AW30-N03B-2-A		
41	Packing Material	Graphite		94	Quantity 1		
42	Packing Type	HT		95	Mat. / Pneu. Conn. Aluminium / 3/8"		
43	Body Gasket	GRPH gasket MESU 1.4401/1.4404		96	Gauges / Filter 1 Pressure gauge, SS / 5 µm		
44	Flow Direction	FTO		97	SOLENOID VALVE		
45	NACE	-		98	Manufacturer / Type /		
46	TRIM			99	Device Type		
47	Valve Coefficient	Cv 190		100	Qty. / Switch. Func. /		
48	Rated Travel	30 mm		101	Material		
49	Seat Bore	100 mm		102	Elec. / Pneu. Conn. /		
50	Stem Ø	16 mm		103	Power Supply		
51	Characteristic	Equal perc.		104	Ex Approval / Cert. /		
52	Noise Reduction	-		105	LIMIT SWITCH		
53	Balanced	-		106	Manufacturer / Type /		
54	Plug Facing	Stellite® faced		107	Device Type		
55	Plug Material	CF3M/F316/F316L/316/316L		108	Quantity		
56	Stem Material	A479 316/A479 316L		109	Material		
57	Seat Facing	Stellited		110	Elec. Conn.		
58	Seat Material	A479 316(L)/A182 F316(L)		111	Ex Approval / Cert. /		
59	Cage Material			112	Contacts		
60	Flow Divider Mat.	-		113	Function		
61	Leakage Class	IV		114	BOOSTER		
62	ACTUATOR			115	Manufacturer / Type SMC / 7019		
63	Manufacturer / Type	SAMSON / 3271		116	Device Type EIL 100-F03L		
64	Size	1000 cm²		117	Quantity 1		
65	Fail Action	Actuator stem extends Fail close		118	Mat. / Pneu. Conn. / G %		
66	Bench/Oper. Range	0.6 to 3.0 bar/2.4 to 3.6 bar		119	ATTENUATION PLATES		
67	Actuator Body Mat.	1.0982 / S460 MC		120	Type		
68	Actuator Style	Pneumatic		121	Size In / Out		
69	Stroke Limit.	-		122	Rating		
70	Membrane Material	NBR		123	Plates Material Qty.		
71	Handwheel	-		124	REMARKS		
72	Set Point / -Range	/		125			
73				126			
74				127			
75	Rev.	Date	Description	Prpd.	Chd.	App.	
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		SAMSON VALVE SPECIFICATION					
		PROJECT SPRINT 18K TPA SPRAY DRYER					
1	Customer	AEC INDUSTRIAL ENGINEERING CO., LTD.			Tag No.	PV-60214	
2	Project No.				Item	2	Qty. 1
3	Unit				Status		
4	PID	325-01-600-PD-03			Quot. No.		Revision 0
5	Line No.				Order No.		Date 26.01.2026
6	PROCESS DATA						
7	Service				Medium	Steam	
8	Phase	vapor			Medium Cust.	Steam (20 Bar G)	
9			Unit	Case 1	Case 2	Case 3	
10	Flow Rate	W	kg/h	1500	1800	2000	
11	Inlet Pressure	p1	bar(g)	16	18	20	
12	Outlet Pressure	p2	bar(g)	15	17.5	18	
13	Inlet Temperature	T1	°C	206	214	240	
14	Spec. Volume	v1	m³/kg	0.117	0.106	0.103	
15	Vapor Pressure	pv	bar(a)				
16	Critical Pressure	pc	bar(a)				
17	Ratio of Specific Heats	γ	-	1.28	1.28	1.29	
18	Compressibility	Z	-				
19	Inlet Viscosity	η	mPas	0.0159	0.0162	0.0174	
20	Outlet Velocity	w	Mach	0.195	0.206	0.234	
21	Valve Coeff. Calculated	Cv	-	19.5	31.4	17.5	
22	Relative Travel	T	%	55.9	73.4	53	
23	SPL(SAMSON Standard values)	LA	dB (A)	60	54	67	
24	DESIGN			77	LINE		
25	Shut-Off Pressure Min	0 bar(g)	Max 24 bar(g)	78	Size / Rating / Pipe Class		
26	t1max		Max 240 °C	79	In NPS 3" / Schedule 40 /		
27	Supply Pressure Min	4 bar(g)	Max bar(g)	80	Out NPS 3" / Schedule 40 /		
28	Pressure Min	bar(a)	Max bar(a)	81	Pipe Inner Dia. / Wallthickness 154,1 mm / 7,1 mm		
29	Temperature Min	°C	Max 250 °C	82	Insu. none		
30	Ambient Temp. Min	°C	Max °C	83	POSITIONER		
31	VALVE BODY / BONNET			84	Manufacturer / Type SAMSON / 3730-1		
32	Manufacturer / Type	SAMSON / 3241		85	Device Type TROVIS 3730-1		
33	Style	Globe		86	Quantity 1		
34	Nominal Size	NPS 2		87	Material Aluminium		
35	Rating	Class 300		88	Elec. / Pneu. Conn. / Connection block G 1/4		
36	Body Material	A216 WCC		89	Input Sig. / Commu. 4 to 20 mA / -		
37	Connection In/Out	flanges		90	Ex Approval / Cert. T3730-1-110 ATEX ia / ATEX		
38	Connection In/Out	RF (Ra 3.2 - 6.3), ASME B16.5		91	AIR SET		
39	Bonnet Type	Insulating section		92	Manufacturer / Type SMC / AW		
40	Bellow Material	-		93	Device Type AW20-N02B-2-D		
41	Packing Material	Graphite		94	Quantity 1		
42	Packing Type	HT		95	Mat. / Pneu. Conn. Aluminium / 1/4"		
43	Body Gasket	GRPH gasket MESU 1.4401/1.4404		96	Gauges / Filter 1 Pressure gauge, SS / 5 µm		
44	Flow Direction	FTO		97	SOLENOID VALVE		
45	NACE	-		98	Manufacturer / Type /		
46	TRIM			99	Device Type /		
47	Valve Coefficient	Cv 47		100	Qty. / Switch. Func. /		
48	Rated Travel	15 mm		101	Material /		
49	Seat Bore	48 mm		102	Elec. / Pneu. Conn. /		
50	Stem Ø	10 mm		103	Power Supply /		
51	Characteristic	Equal perc.		104	Ex Approval / Cert. /		
52	Noise Reduction	-		105	LIMIT SWITCH		
53	Balanced	-		106	Manufacturer / Type /		
54	Plug Facing	Stellite® faced		107	Device Type /		
55	Plug Material	CF3M/F316/F316L/316/316L		108	Quantity /		
56	Stem Material	A479 316/A479 316L		109	Material /		
57	Seat Facing	Stellited		110	Elec. Conn. /		
58	Seat Material	A182 F6a Cl. 2 / A 276 410 T		111	Ex Approval / Cert. /		
59	Cage Material	-		112	Contacts /		
60	Flow Divider Mat.	-		113	Function /		
61	Leakage Class	IV		114	BOOSTER		
62	ACTUATOR			115	Manufacturer / Type /		
63	Manufacturer / Type	SAMSON / 3277		116	Device Type /		
64	Size	350 cm²		117	Quantity /		
65	Fail Action	Fail close		118	Mat. / Pneu. Conn. /		
66	Bench/Oper. Range	2.1 to 3.3 bar/2.1 to 3.3 bar		119	ATTENUATION PLATES		
67	Actuator Body Mat.	1.0976 / 1.0982		120	Type /		
68	Actuator Style	Pneumatic		121	Size In / Out /		
69	Stroke Limit.	-		122	Rating /		
70	Membrane Material	NBR		123	Plates Material Qty. /		
71	Handwheel	-		124	REMARKS		
72	Set Point / -Range	/		125			
73				126			
74				127			
75	Rev.	Date	Description	Prpd.			
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